Day 40 — TNPSC Group 2A Study Material

Part A – General Studies: Nature of Indian Economy, Five-Year Plans, Planning Commission and NITI Aayog

READING MATERIAL

Covers India's economic system (a 'mixed economy' blending public and private sectors), the historical Five-Year Plan models (1951-2017), and their replacement by NITI Aayog in 2015. Real exam evidence rewards precise recall of exact plan-era themes and personnel, not just broad concepts.

Q: What kind of economic system does India have?

A: A 'mixed economy' — combining elements of both a public sector (government-owned/controlled enterprises and planning) and a private sector (market-driven enterprise), rather than being purely socialist/state-controlled or purely capitalist/free-market.

Q: When did Five-Year Plans begin in India, and what replaced this planning model, and when?

A: The First Five-Year Plan began in 1951, under the Planning Commission (established 1950). After 12 Five-Year Plans (the 12th ran 2012-2017), the model was replaced by NITI Aayog (National Institution for Transforming India), established in January 2015 — shifting from centralised 5-year planning to a more flexible advisory/cooperative-federalism model.

Q: Real exam: What was the actual, precisely-worded main target/theme of the Eleventh Five Year Plan (2007-2012)?

A: 'Rapid, sustainable, and more inclusive growth' — confirmed correct by a real exam question, which tested this exact phrasing AGAINST a similar-sounding but imprecise alternative ('equitable development' alone), which was marked incorrect as the plan's stated theme.

Q: Real exam: What was the Gandhian Plan (1944, authored by S.N. Agarwala), and what was its REAL core objective?

A: A Gandhian-inspired alternative economic vision proposed during the freedom struggle era. Its real core objective, per a real exam question, was to raise BOTH the material AND the cultural level of the Indian masses — broader than narrower-sounding distractor options like 'agricultural development alone' or 'growth of cottage/village industries alone' (those were elements/means within the plan, not its overall stated objective).

Q: Real exam: Who was the LAST Deputy Chairman of the Planning Commission of India, before it was replaced?

A: Montek Singh Ahluwalia — confirmed by a real exam question, distinguishing him from Manmohan Singh, Arun Jaitley, and others offered as distractors.

Q: What does NITI Aayog do, including a notable recent function?

A: NITI Aayog serves as the government's policy think-tank, fostering cooperative federalism and replacing the old centralised planning model. A notable recent function: it publishes the annual SDG (Sustainable Development Goals) India Index, ranking States/UTs on their progress toward the UN's Sustainable Development Goals — note that specific year-by-year rankings (e.g., which state ranks where in the 2023-24 edition) change annually and shouldn't be memorized as permanent facts.

Q: Real exam: What replaced the MRTP Act, 1969 (already noted in the Consumer Protection topic), and from when?

A: The Competition Act, 2002 replaced the MRTP Act, 1969 — taking effect from 2009 (the repeal and full transition took a few years after the Act was passed) — confirmed by a real exam question.

MOCK TEST (WITH ANSWERS)

1. India's economy is best described as a:

A. Purely socialist economy

B. Purely capitalist economy

C. Mixed economy ✓

D. Barter economy

Explanation: Mixed economy. [easy, authored]

2. NITI Aayog replaced the Planning Commission in which year?

A. 2009

B. 2012

C. 2015 ✓

D. 2017

Explanation: 2015. [medium, authored]

3. What was the official theme/main target of the Eleventh Five Year Plan (2007-2012)?

A. Equitable development alone

B. Rapid, sustainable, and more inclusive growth ✓

C. Self-sufficiency in food grains alone

D. Privatisation of all public sector units

Explanation: Rapid, sustainable, and more inclusive growth. [hard, question-bank-2024]

4. The Gandhian Plan (1944), authored by S.N. Agarwala, had which real core objective?

A. Agricultural development only

B. Growth of cottage and village industries only

C. To achieve absolute equality only

D. To raise both the material and cultural level of Indian masses ✓

Explanation: Raise both the material and cultural level of Indian masses. [hard, question-bank-2017]

5. Who was the last Deputy Chairman of the Planning Commission of India?

A. Manmohan Singh

B. Narendra Modi

C. Arun Jaitley

D. Montek Singh Ahluwalia ✓

Explanation: Montek Singh Ahluwalia. [hard, question-bank-2017]

6. NITI Aayog publishes an annual index ranking States/UTs on progress toward the UN's Sustainable Development Goals, called the:

A. Human Development Index

B. Ease of Doing Business Index

C. SDG India Index ✓

D. Global Competitiveness Index

Explanation: SDG India Index. [medium, question-bank-2025]

7. The MRTP Act, 1969 was repealed and replaced by the Competition Act, 2002, with effect from:

A. 2002

B. 2009 ✓

C. 2013

D. 2016

Explanation: 2009. [hard, question-bank-2017]

8. The First Five-Year Plan of India began in:

A. 1947

B. 1950

C. 1951 ✓

D. 1956

Explanation: 1951. [medium, authored]

Part B – Aptitude & Mental Ability: Area

READING MATERIAL

Area is 1 of 10 named Aptitude sub-skills. Real exam evidence shows a strong preference for applied/real-world framing (fencing cost, carpet cost, crossroads, a grazing horse) over bare formula plug-ins — and frequently combines area with a real-world cost-per-unit calculation.

Q: Method — area of a triangle from coordinates: Find the area of the triangle with vertices (0,0), (2,0), (0,2).

A: When two sides lie along the axes (a right triangle at the origin), area = ½ × base × height = ½ × 2 × 2 = 2 sq. units. (The general coordinate formula $\frac{1}{2}|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)|$ also works, but recognizing the right-angle-at-origin shortcut is faster.)

Q: Real exam: A rectangular ground 80m × 60m has two equal-width cross roads (one parallel to each side). Their combined area is 675 sq.m. Find the road width.

A: If width=w, the two roads' combined area = $80w + 60w - w^2$ (subtracting the overlapping square where they cross, counted twice otherwise) = $140w - w^2$. Set equal to 675: $w^2 - 140w + 675 = 0$. Solving gives w=5 (the other root, 135, is too large to be physically valid for an 80×60 ground).

Q: Real exam: Find the perimeter of a semicircle of radius 28cm.

A: Perimeter of a semicircle = πr + 2r (the curved half plus the straight diameter edge). Using π≈22/7: (22/7×28) + (2×28) = 88 + 56 = 144cm. (Common mistake: forgetting to add the straight diameter edge, which would wrongly give just 88cm.)

Q: Real exam: A horse is tethered by a 14m rope to one corner of a 60m×42m rectangular field. How much area is left UNGRAZED?

A: The horse can graze a quarter-circle of radius 14m (since it's tied at a corner, with two field-edges as boundaries): grazed area = $\frac{1}{4}\times\pi\times14^2 = \frac{1}{4}\times(22/7)\times196$ = 154 sq.m. Total field area = 60×42 = 2520 sq.m. Ungrazed = 2520−154 = 2366 sq.m.

Q: Real exam: Area of a rhombus with diagonals 6cm and 8cm?

A: Area of rhombus = $\frac{1}{2} \times d_1 \times d_2$ = ½×6×8 = 24 cm^2.

Q: Real exam: If fencing a circular park at ₹5/metre costs ₹1,100 total, find the park's radius.

A: Total fencing length (= circumference) = ₹1,100 ÷ ₹5/m = 220m. Circumference = 2πr → 220 = 2×(22/7)×r → r = 220×7/44 = 35m.

Q: Real exam: Find the cost of carpeting a room 13m long, 9m wide, using a carpet 75cm wide, at Rs.8 per metre (of carpet length).

A: Room area = 13×9 = 117 m^2. Carpet width = 0.75m, so length of carpet needed = Room area ÷ carpet width = 117/0.75 = 156m. Cost = 156 × Rs.8 = Rs.1,248.

Q: Real exam (related, surface area of a 3D shape): The curved surface area of a right circular cylinder of height 7cm is 66 cm^2. Find its diameter.

A: CSA = 2πrh → 66 = 2×(22/7)×r×7 = 44r → r=1.5cm → diameter = 3cm.

MOCK TEST (WITH ANSWERS)

1. Find the area of the triangle with vertices (0,0), (5,0), (0,4).

A. 10 sq. units ✓

B. 20 sq. units

C. 9 sq. units

D. 18 sq. units

Explanation: Right triangle at origin: area = ½×5×4 = 10 sq. units. [easy, authored]

2. Find the perimeter of a semicircle of radius 21cm (use π≈22/7).

A. 66 cm

B. 108 cm ✓

C. 132 cm

D. 96 cm

Explanation: Perimeter = πr+2r = (22/7×21)+(2×21) = 66+42 = 108cm. [medium, authored]

3. Find the area of a rhombus with diagonals 10cm and 12cm.

A. 60 cm^2 ✓

B. 120 cm^2

C. 30 cm^2

D. 100 cm^2

Explanation: Area = $\frac{1}{2} \times 10 \times 12 = 60$ cm^2. [easy, authored]

4. Fencing a circular garden at ₹4 per metre costs ₹880 in total. Find the garden's radius (use $\pi \approx 22/7$).

A. 28 m

B. 35 m ✓

C. 22 m

D. 14 m

Explanation: Circumference = 880/4 = 220m. r = 220×7/(2×22) = 1540/44 = 35m. [medium, authored]

5. Find the cost of carpeting a room 10m long and 8m wide with a carpet 80cm wide, at Rs.10 per metre.

A. Rs.800

B. Rs.1,000 ✓

C. Rs.1,200

D. Rs.960

Explanation: Room area = 80 m^2. Carpet length needed = 80/0.8 = 100m. Cost = 100×Rs.10 = Rs.1,000. [medium, authored]

6. A horse is tethered by a 7m rope to a corner of a 20m×15m rectangular field. Find the grazed area (use $\pi \approx 22/7$).

A. 38.5 m^2 ✓

B. 44 m^2

C. 49 m^2

D. 77 m^2

Explanation: Quarter circle of radius 7m: $\frac{1}{4} \times (22/7) \times 49 = 38.5$ m^2. [medium, authored]

Part C – General English: Prose: "From Zero to Infinity" — Biography of Srinivasa Ramanujan

READING MATERIAL

A biographical account of the mathematical genius Srinivasa Ramanujan, opening with a memorable classroom anecdote about dividing zero by zero, and tracing his early education, self-taught brilliance, struggles in formal college, and first job.

Q: What classroom anecdote opens the biography, and what question did young Ramanujan ask that amused his classmates?

A: During an ARITHMETIC class, Ramanujan asked: "Can zero divided by zero give one?" — a question that amused his classmates but reflected his unusually deep mathematical curiosity even as a child.

Q: What was the teacher's explanation for zero divided by zero, and who is credited with first proving this?

A: The teacher explained that zero divided by zero is INFINITY. The biography credits the ancient mathematician BHASKARA with proving this.

Q: Where was Srinivasa Ramanujan born, and what was his father's profession?

A: He was born in Erode (Tamil Nadu). His father worked as a CLERK in a cloth shop.

Q: At what age did Ramanujan study S.L. Loney's Trigonometry, and which book with thousands of theorems later inspired him?

A: At age 13. He was later deeply inspired by a book called 'Synopsis of Elementary Results' (in Pure and Applied Mathematics), containing 4,865 theorems.

Q: What name became famous for Ramanujan's personal mathematical notebooks, and why did he struggle in formal college despite his genius?

A: They became famous as 'Ramanujan's Frayed Notebooks'. Despite his mathematical brilliance, he FAILED college in History, English, and Physiology — subjects outside his singular passion for mathematics, since he devoted almost all his attention to math and neglected the rest of the curriculum.

Q: Who gave Ramanujan his first job, what was the position, and what was his initial pay?

A: Francis Spring gave him a job as a CLERK at the Madras Port Trust. His initial pay was just ₹25 per month.

MOCK TEST (WITH ANSWERS)

1. What subject was being taught when young Ramanujan asked his famous question about zero?

A. Algebra

B. Arithmetic ✓

C. Geometry

D. Trigonometry

Explanation: Arithmetic. [easy, web-research]

2. What question did Ramanujan ask that amused his classmates?

A. Why is 1 ÷ 1 = 1?

B. Can zero divided by zero give one? ✓

C. Why is 3 ÷ 3 = 1?

D. Can 1000 be divided?

Explanation: Can zero divided by zero give one? [medium, web-research]

3. According to the teacher's explanation in the biography, what does zero divided by zero equal?

A. Zero

B. One

C. Undefined

D. Infinity ✓

Explanation: Infinity. [medium, web-research]

4. Who is credited in the biography with first proving that zero divided by zero is infinity?

A. Ramanujan

B. Aryabhatta

C. Bhaskara ✓

D. G.H. Hardy

Explanation: Bhaskara. [hard, web-research]

5. Where was Srinivasa Ramanujan born?

A. Erode ✓

B. Kumbakonam

C. Chennai

D. Madurai

Explanation: Erode. [medium, web-research]

6. Ramanujan's father worked as a:

A. Professor

B. Priest

C. Cloth shop clerk ✓

D. Farmer

Explanation: Cloth shop clerk. [medium, web-research]

7. At what age did Ramanujan study S.L. Loney's Trigonometry?

A. 10

B. 12

C. 13 ✓

D. 15

Explanation: 13. [hard, web-research]

8. Which book, containing 4,865 theorems, deeply inspired Ramanujan?

A. Number Theory

B. Synopsis of Elementary Results ✓

C. Trigonometry of India

D. Ramanujan's Notebook

Explanation: Synopsis of Elementary Results. [hard, web-research]

9. Ramanujan's personal mathematical notebooks later became famous as:

A. Ramanujan's Math Book

B. Infinity Pages

C. Ramanujan's Frayed Notebooks ✓

D. Ramanujan's Scripts

Explanation: Ramanujan's Frayed Notebooks. [medium, web-research]

10. Which subjects did Ramanujan fail in college, despite his mathematical genius?

A. History, Geography, Chemistry

B. Math, English, History

C. History, English, Physiology ✓

D. English, Biology, Physics

Explanation: History, English, Physiology. [hard, web-research]

11. Francis Spring gave Ramanujan his first job as a:

A. Typist

B. Clerk at Madras Port Trust ✓

C. Mathematics teacher

D. Railway accountant

Explanation: Clerk at Madras Port Trust. [medium, web-research]

12. What was Ramanujan's initial monthly pay as a clerk?

A. ₹75

B. ₹50

C. ₹30

D. ₹25 ✓

Explanation: ₹25. [hard, web-research]